

If you want to **remove everything you have done for this Cowrie setup and start completely fresh**, the easiest method is to delete the Cowrie project folder and its virtual environment.

You are currently inside ~/cowrie, so first exit the virtual environment and move to your home directory:

```
deactivate
```

```
cd ~
```

```
pwd
```

It should show:

```
/home/kali
```

Now check what Cowrie-related folders you have:

```
ls -la | grep -i cowrie
```

If you see cowrie and possibly cowrie_backup, remove them:

```
rm -rf ~/cowrie
```

```
rm -rf ~/cowrie_backup
```

You also previously had an old Cowrie folder in the Trash. If you want to remove **only that Cowrie Trash copy** too:

```
rm -rf ~/.local/share/Trash/files/cowrie
```

Now verify:

```
ls -la ~ | grep -i cowrie
```

```
find /home/kali -maxdepth 3 -iname "*cowrie*" 2>/dev/null
```

If these return nothing relevant, your previous Cowrie setup has been cleared.

Don't run sudo rm -rf for this. Your Cowrie installation is under /home/kali, so normal rm should be sufficient.

Cowrie Honeypot — Complete Commands From Start to End

1. Update Kali Linux

```
sudo apt update
```

Optional full upgrade:

```
sudo apt upgrade -y
```

2. Install required packages

```
sudo apt install -y git python3 python3-venv python3-dev libssl-dev libffi-dev build-essential
```

Check Python:

```
python3 --version
```

Your Python **3.13.12** is suitable for continuing.

3. Go to home directory

```
cd ~
```

Check:

```
pwd
```

It should show:

```
/home/kali
```

4. Download Cowrie

```
git clone https://github.com/cowrie/cowrie.git
```

Enter the directory:

```
cd cowrie
```

Check files:

```
ls
```

You should see files/directories similar to:

bin

docs

etc

src

var

requirements.txt

pyproject.toml

README.rst

5. Create Python virtual environment

```
python3 -m venv cowrie-env
```

Activate it:

```
source cowrie-env/bin/activate
```

Now the terminal should start with:

(cowrie-env)

For example:

```
(cowrie-env) kali@kali:~/cowrie$
```

6. Upgrade pip

```
python -m pip install --upgrade pip
```

7. Install Cowrie

```
python -m pip install -e .
```

Wait until installation completes.

8. Check Cowrie installation

```
which cowrie
```

You should get something similar to:

```
/home/kali/cowrie/cowrie-env/bin/cowrie
```

Then:

```
cowrie --help
```

9. Initialize Cowrie

```
cowrie init
```

Now check:

```
ls -la etc/
```

You should have:

```
cowrie.cfg
```

You can also check:

```
ls -la var/
```

10. Open Cowrie configuration

```
nano etc/cowrie.cfg
```

For the first experiment, you can normally keep the default SSH configuration.

To save in Nano:

```
Ctrl + O
```

Press:

```
Enter
```

Then exit:

```
Ctrl + X
```

11. Start the Cowrie honeypot

Make sure you are in the Cowrie directory:

```
cd ~/cowrie
```

Start:

```
cowrie start
```

You may see:

```
Starting cowrie: [twistd ... cowrie]...
```

That means Cowrie has been launched.

12. Check Cowrie status

```
cowrie status
```

13. Check SSH honeypot port

Cowrie commonly uses **2222** for its SSH honeypot.

```
ss -lntp | grep 2222
```

You should see 2222 in LISTEN state.

14. Check startup logs

```
tail -30 var/log/cowrie/cowrie.log
```

Look for errors and confirm that the SSH service started.

15. Monitor Cowrie live

Keep the first terminal running.

Use:

```
tail -f var/log/cowrie/cowrie.log
```

This will continuously display activity.

To stop viewing the log:

Ctrl + C

This only stops tail; **Cowrie continues running.**

Testing the Honeypot

16. Open a second terminal

Open another Kali Terminal.

Test Cowrie using:

```
ssh -p 2222 root@127.0.0.1
```

 or open putty type ip address port 2222

Or:

```
ssh -p 2222 root@localhost
```

17. Accept the SSH fingerprint

The first time you may see:

Are you sure you want to continue connecting (yes/no/[fingerprint])?

Enter:

yes

It will then ask for a password.

18. Check Cowrie's test credential configuration

If you need to inspect the credential database, return to the Cowrie terminal and run:

```
cd ~/cowrie
```

Find the user database:

```
find . -name "userdb*"
```

If you have:

```
etc/userdb.txt
```

view it:

```
cat etc/userdb.txt
```

Do **not** put your real Kali password in the Cowrie credential database. Use only fake lab credentials.

19. Login to the honeypot

Again:

```
ssh -p 2222 root@127.0.0.1
```

Enter a configured test password.

After successful authentication, you'll be inside Cowrie's simulated shell.

20. Test some commands

Try:

```
whoami
```

Then:

```
pwd
```

Then:

```
ls
```

Then:

```
uname -a
```

Then:

```
cat /etc/passwd
```

You can also try:

```
ip addr
```

These are being entered into the **Cowrie honeypot environment**, not your actual Kali shell.

21. Exit the honeypot

```
exit
```

Monitor Captured Activity

22. Return to the Cowrie terminal

```
cd ~/cowrie
```

23. View recent Cowrie logs

```
tail -50 var/log/cowrie/cowrie.log
```

24. Monitor logs live

```
tail -f var/log/cowrie/cowrie.log
```

Now if another computer connects to your Cowrie honeypot, you can watch the activity appear.

Press:

Ctrl + C

when you want to stop live viewing.

25. View JSON logs

```
tail -50 var/log/cowrie/cowrie.json
```

For live monitoring:

```
tail -f var/log/cowrie/cowrie.json
```

26. Search login activity

```
grep -i "login" var/log/cowrie/cowrie.log
```

27. Search password attempts

```
grep -i "password" var/log/cowrie/cowrie.log
```

This is useful for your experiment because Cowrie can record attempted authentication activity.

Testing from Another Computer

If your Kali VM IP is, for example:

192.168.1.50

another machine on your **authorized lab network** can test:

```
ssh -p 2222 root@192.168.1.50
```

Find your Kali IP using:

```
ip addr
```

or:

```
hostname -I
```

For VMware, make sure the networking configuration allows the two lab machines to communicate. Keep this isolated from sensitive systems.

Restart Cowrie

If you modify:

etc/cowrie.cfg

restart Cowrie:

cowrie restart

Then:

cowrie status

Stop Cowrie

28. Stop the honeypot

cowrie stop

29. Confirm it stopped

cowrie status

You can also check:

ss -lntp | grep 2222

There should no longer be a Cowrie listener on 2222.

30. Exit virtual environment

deactivate

Your (cowrie-env) prefix should disappear.

Next Time You Want to Run Cowrie

You **do not** reinstall everything.

Just open Terminal and run:

cd ~/cowrie

Activate:

source cowrie-env/bin/activate

Start:

cowrie start

Check:

cowrie status

Monitor:

tail -f var/log/cowrie/cowrie.log

When finished:

cowrie stop

Then:

deactivate

Short Practical Command Sequence

For your lab record, the essential installation sequence is:

sudo apt update

sudo apt install -y git python3 python3-venv python3-dev libssl-dev libffi-dev build-essential

cd ~

git clone https://github.com/cowrie/cowrie.git

cd cowrie

python3 -m venv cowrie-env

source cowrie-env/bin/activate

python -m pip install --upgrade pip

python -m pip install -e .

cowrie init

```
cowrie start
```

```
cowrie status
```

```
ss -lntp | grep 2222
```

Second Terminal:

```
ssh -p 2222 root@127.0.0.1
```

Monitor:

```
cd ~/cowrie
```

```
tail -f var/log/cowrie/cowrie.log
```

Finish experiment:

```
cowrie stop
```

```
deactivate
```

That gives you the complete flow: **Install → Configure → Start Honeypot → Test SSH → Capture Activity → Analyze Logs → Stop Honeypot.**